Arithmetic & Number Sense

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Powers and roots

Evaluate $2^3 \times 2^2$ using the laws of exponents.

$2^3 \times 2^2 = 2^{3+2} = 2^5 = 32$.

2 [Foundational] • Target: Working with fractions

Which is larger, $\frac{5}{7}$ or $\frac{4}{5}$?

Cross-multiply: $5 \times 5 = 25$ and $4 \times 7 = 28$. Since $28 > 25$, $\frac{4}{5}$ is larger.

3 [Foundational] • Target: Place value and order of magnitude

Write $4{,}500{,}000$ in the form $a \times 10^n$ with $1 \le a < 10$.

$4{,}500{,}000 = 4.5 \times 10^6$.

4 [Foundational] • Target: Place value and order of magnitude

Which is greater, 6×10^4 or $58{,}000$?

$6 \times 10^4 = 60{,}000$, and $60{,}000 > 58{,}000$, so 6×10^4 is greater.

5 [Foundational] • Target: Place value and order of magnitude

Order from least to greatest: 0.6, $\frac{2}{3}$, 0.65.

$\frac{2}{3} \approx 0.667$. Ordered: $0.6 < 0.65 < \frac{2}{3}$.

6 [Foundational] • Target: Mixed numbers and improper fractions

Convert $4\frac{2}{5}$ to an improper fraction.

$4\frac{2}{5} = \frac{4 \times 5 + 2}{5} = \frac{22}{5}$.

7 [Foundational] • Target: Mixed numbers and improper fractions

Convert $\frac{47}{6}$ to a mixed number.

$47 \div 6 = 7$ remainder 5, so $\frac{47}{6} = 7\frac{5}{6}$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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8 [Intermediate] • Target: Factors, multiples, GCD and LCM

Find the greatest common divisor (GCD) of 48 and 60.

$48 = 2^4 \times 3$ and $60 = 2^2 \times 3 \times 5$. Common factors: $2^2 \times 3 = 12$.

9 [Intermediate] • Target: Factors, multiples, GCD and LCM

Find the least common multiple (LCM) of 8 and 12.

$8 = 2^3$, $12 = 2^2 \times 3$. LCM $= 2^3 \times 3 = 24$.

10 [Intermediate] • Target: Factors, multiples, GCD and LCM

List all the factors of 36, and state how many there are.

$1, 2, 3, 4, 6, 9, 12, 18, 36$ — that is 9 factors.

11 [Intermediate] • Target: Factors, multiples, GCD and LCM

Is 91 a prime number? Justify your answer.

$91 = 7 \times 13$, so it has factors other than 1 and itself. 91 is not prime.

12 [Intermediate] • Target: Factors, multiples, GCD and LCM

Find the greatest common factor of 84 and 126.

$84 = 2^2 \times 3 \times 7$, $126 = 2 \times 3^2 \times 7$. Common factors: $2 \times 3 \times 7 = 42$.

13 [Intermediate] • Target: Divisibility rules

Without dividing, determine whether $4{,}578$ is divisible by 3.

Digit sum: $4 + 5 + 7 + 8 = 24$, which is divisible by 3 (since $24 = 3 \times 8$). So $4{,}578$ is divisible by 3.

14 [Intermediate] • Target: Divisibility rules

Is $2{,}340$ divisible by 9?

Digit sum: $2 + 3 + 4 + 0 = 9$, which is divisible by 9. So $2{,}340$ is divisible by 9.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 3.01 Algebra Essentials

Estimate 198×51 by rounding each number to the nearest ten, then find the exact product.

Rounded: $200 \times 50 = 10{,}000$. Exact: $198 \times 51 = 10{,}098$ — close to the estimate.

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16 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 4.01 Geometry & Measurement

Round 2.4863 to two decimal places.

The third decimal digit is 6, so round up: 2.49.

17 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 5.01 Quantitative Comparisons

Estimate $\sqrt{50}$ to the nearest whole number.

$7^2 = 49$ and $8^2 = 64$; since 50 is closest to 49, the estimate is 7 (actual value ≈ 7.07).

18 [Challenging] • Target: Estimation and rounding • Cross-Lesson: 6.01 Sequences & Patterns

A shipment weighs approximately $2{,}850$ kg. Round this to the nearest hundred kilograms.

The tens digit is 5, so round up: $2{,}900$ kg.

19 [Challenging] • Target: Number properties • Cross-Lesson: 7.01 Data Analysis & Probability

If n is an even integer, is $n^2 + n$ always even, always odd, or can it be either?

Write $n = 2k$. Then $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$, which is always even.

20 [Challenging] • Target: Number properties • Cross-Lesson: 8.01 Word Problems & Reasoning

Find the remainder when 47 is divided by 6.

$6 \times 7 = 42$, and $47 - 42 = 5$. The remainder is 5.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: MCQ — Exponents and roots

Use the laws of exponents to evaluate $3^2 \times 3^4$.

a. 243

b. 81

c. 729 ✓

d. 27

$3^2 \times 3^4 = 3^{2+4} = 3^6 = 729$.

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22 [Foundational] • Target: MCQ — Comparing fractions and decimals

Which value is largest: the point shown on the number line, $\frac{3}{5}$, or 0.55?

a. **The marked point (0.7)** ✓

b. $\frac{3}{5}$

c. They are all equal

d. 0.55

The marked point is at 0.7. Since $\frac{3}{5} = 0.6$ and $0.55 < 0.6 < 0.7$, the marked point is largest.

23 [Foundational] • Target: MCQ — Decimal operations

The number line marks the value of $9.2 - 3.45$. Where does it land?

a. 5.75 ✓

b. 6.25

c. 5.85

d. 5.65

Align decimal points: $9.20 - 3.45 = 5.75$.

24 [Foundational] • Target: MCQ — Scientific notation & place value

The bar chart shows four city populations. Which one equals 7×10^3?

a. $7{,}000$ ✓

b. $6{,}800$

c. $6{,}500$

d. $7{,}300$

$7 \times 10^3 = 7{,}000$, matching the second bar exactly.

25 [Foundational] • Target: MCQ — Scientific notation & place value

Evaluate 3.4×10^4 as an ordinary number.

a. $3{,}400{,}000$

b. $3{,}400$

c. $340{,}000$

d. $34{,}000$ ✓

Move the decimal point 4 places right: $3.4 \times 10^4 = 34{,}000$.

26 [Foundational] • Target: MCQ — Scientific notation & place value

In the number $528{,}437$, what is the place value of the digit 8?

a. Hundred thousands

b. **Thousands** ✓

c. Ten thousands

d. Hundreds

Counting from the right: 7 (ones), 3 (tens), 4 (hundreds), 8 (thousands).

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27 [Foundational] • Target: MCQ — Absolute value & signed numbers

Two of the four points on the number line represent a pair of numbers that are equal in absolute value but opposite in sign. Which pair?

a. -2 and 2

b. -4 **and** 4 ✓

c. -2 and 4

d. -4 and 2

-4 and 4 are the same distance from 0 (4 units) on opposite sides, so they are opposites.

28 [Foundational] • Target: MCQ — GCD, LCM & factors

How many factors does 60 have?

a. 6

b. 8

c. 12 ✓

d. 10

$60 = 2^2 \times 3 \times 5$. Number of factors $= (2+1)(1+1)(1+1) = 12$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: MCQ — GCD, LCM & factors

The bar chart shows a gardener's bulb counts. They want to plant them in identical rows with the largest possible number of bulbs per row, each row containing only one type of flower. How many bulbs per row?

a. 4

b. 24

c. 6

d. 12 ✓

The largest common row size is the GCD of 36 and 48: $36 = 2^2 \times 3^2$, $48 = 2^4 \times 3$, GCD $= 2^2 \times 3 = 12$.

30 [Intermediate] • Target: MCQ — Divisibility rules

Without dividing, is $5{,}481$ divisible by 3?

a. Cannot tell without dividing

b. **Yes** ✓

c. No

d. Only by 9, not 3

Digit sum: $5 + 4 + 8 + 1 = 18$, divisible by 3 (and by 9), so $5{,}481$ is divisible by 3.

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31 [Intermediate] • Target: MCQ — Divisibility rules

The bar chart shows four candidate numbers. Which one is divisible by both 5 and 6?

a. 690 ✓

b. 650

c. 685

d. 696

Divisible by 5: ends in 0 or 5. Divisible by 6: even and digit sum divisible by 3. Only 690 satisfies both.

32 [Intermediate] • Target: MCQ — Estimation & rounding

Round 147.362 to the nearest tenth.

a. 147.3

b. 150

c. 147.36

d. 147.4 ✓

The hundredths digit is 6, so round up: 147.4.

33 [Intermediate] • Target: MCQ — Estimation & rounding

The number line shows the value x rounded to the nearest whole number. If $x = 6.7$, where does the rounded value land?

a. 6

b. 6.5

c. 8

d. 7 ✓

6.7 is closer to 7 than to 6 (since $.7 > .5$), so it rounds to 7.

34 [Intermediate] • Target: MCQ — Estimation & rounding

Estimate 312×48 by rounding each factor to the nearest ten, then compare to the exact product $14{,}976$.

a. Estimate $310 \times 50 = 15{,}500$, far from the exact value

b. Estimate $300 \times 50 = 15{,}000$, close to the exact value

c. Estimate $320 \times 40 = 12{,}800$, close to the exact value

d. **Estimate $310 \times 50 = 15{,}500$, close to the exact value** ✓

312 rounds to 310 and 48 rounds to 50: $310 \times 50 = 15{,}500$, reasonably close to $14{,}976$.

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35 [Intermediate] • Target: MCQ — Number properties

If n is an odd integer, is $n^2 + n$ always even, always odd, or can it be either?

a. Can be either

b. Depends on the sign of n

c. Always odd

d. **Always even ✓**

Write $n = 2k + 1$. Then $n^2 + n = n(n + 1)$, a product of two consecutive integers — one of which is always even, so the product is always even.

36 [Intermediate] • Target: MCQ — Number properties

The number line marks 85 between consecutive multiples of 9. Find the remainder when 85 is divided by 9.

a. 3

b. 4 ✓

c. 5

d. 76

$9 \times 9 = 81$, and $85 - 81 = 4$. The remainder is 4.

37 [Intermediate] • Target: MCQ — Decimal operations

A shipment of 277 pencils is divided as equally as possible among 15 boxes. How many pencils does each box receive, and how many are left over?

a. $18, 7$ ✓

b. $18, 8$

c. $7, 18$

d. $19, 7$

$277 \div 15 = 18$ remainder 7 (since $15 \times 18 = 270$). Each receives 18, with 7 left over.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: MCQ — Scientific notation & place value • Cross-Lesson: 6.01 Sequences & Patterns

A shipment of 310 books is divided as equally as possible among 15 shelves. How many books does each shelf receive, and how many are left over?

a. $20, 11$

b. $21, 10$

c. $10, 20$

d. $20, 10$ ✓

$310 \div 15 = 20$ remainder 10 (since $15 \times 20 = 300$). Each receives 20, with 10 left over.

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39 [Challenging] • Target: MCQ — Mixed numbers & improper fractions • Cross-Lesson: 7.01 Data Analysis & Probability

A shipment of 147 stickers is divided as equally as possible among 15 students. How many stickers does each student receive, and how many are left over?

a. $9, 12$ ✓

b. $9, 13$

c. $10, 12$

d. $12, 9$

$147 \div 15 = 9$ remainder 12 (since $15 \times 9 = 135$). Each receives 9, with 12 left over.

40 [Challenging] • Target: MCQ — Absolute value & signed numbers • Cross-Lesson: 8.01 Word Problems & Reasoning

A shipment of 336 chairs is divided as equally as possible among 23 rows. How many chairs does each row receive, and how many are left over?

a. $16, 13$

b. $14, 14$ ✓

c. $15, 14$

d. $14, 15$

$336 \div 23 = 14$ remainder 14 (since $23 \times 14 = 322$). Each receives 14, with 14 left over.

41 [Challenging] • Target: MCQ — GCD, LCM & factors • Cross-Lesson: 2.01 Ratios, Proportions & Percentages

A shipment of 253 cups is divided as equally as possible among 14 bags. How many cups does each bag receive, and how many are left over?

a. $1, 18$

b. $19, 1$

c. $18, 2$

d. $18, 1$ ✓

$253 \div 14 = 18$ remainder 1 (since $14 \times 18 = 252$). Each receives 18, with 1 left over.

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42 [Challenging] • Target: MCQ — Divisibility rules • Cross-Lesson: 3.01 Algebra Essentials

[illegible] $x^2 - 6x + k = 0$[illegible] [illegible] k [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 6 [illegible] [illegible] [illegible] [illegible] [illegible] 5[illegible] [illegible] [illegible] [illegible] k [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] We have the equation $x^2 - 6x + k = 0$, where k is a constant. If the sum of the roots of this quadratic equation is 6 and the product of its roots is 5, what is the value of k, and what is the larger of the two roots?

a. $k = 5$[illegible] [illegible] [illegible] 5 ✓

b. $k = 5$[illegible] [illegible] [illegible] 1

c. $k = 6$[illegible] [illegible] [illegible] 5

d. $k = 6$[illegible] [illegible] [illegible] 1

[illegible] [illegible] [illegible] [illegible] $ax^2 + bx + c = 0$[illegible] [illegible] [illegible] [illegible] [illegible] $-b/a$ [illegible] [illegible] [illegible] [illegible] c/a. [illegible] [illegible] $x^2 - 6x + k = 0$[illegible] [illegible] $a = 1$[illegible] $b = -6$[illegible] [illegible] $c = k$. [illegible] [illegible] [illegible] [illegible] [illegible] 6. [illegible] [illegible] $-b/a = -(-6)/1 = 6$[illegible] [illegible] [illegible] [illegible] [illegible]. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 5. [illegible] [illegible] $c/a = k/1 = k$. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] 5[illegible] [illegible] $k = 5$. [illegible] [illegible] [illegible] $k = 5$[illegible] [illegible] [illegible] [illegible] [illegible] $x^2 - 6x + 5 = 0$. [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible]: $(x - 1)(x - 5) = 0$. [illegible] [illegible] [illegible] [illegible].

43 [Challenging] • Target: MCQ — Estimation & rounding • Cross-Lesson: 4.01 Geometry & Measurement

[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] How many 2-digit even numbers have distinct digits?

a. 40

b. **41** ✓

c. 45

d. 49

[illegible] [illegible] [illegible] [illegible] [illegible] 45. [illegible] [illegible] [illegible] [illegible]: 22, 44, 66, 88 (4 [illegible]). $45 - 4 = 41$.

44 [Challenging] • Target: MCQ — Number properties • Cross-Lesson: 5.01 Quantitative Comparisons

[illegible] [illegible] [illegible] [illegible] [illegible] $y = f(x)$. [illegible] [illegible] [illegible] $f(2)$[illegible] [illegible] [illegible] [illegible] The graph above shows the function $y = f(x)$. What is the approximate value of $f(2)$?

a. 2

b. 3

c. **4** ✓

d. 5

[illegible] [illegible] $f(2)$[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $x = 2$. [illegible] $x = 2$[illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] [illegible] $y = 4$. [illegible] $f(2) = 4$. To find the value of $f(2)$, we locate the point on the graph where $x = 2$. At $x = 2$, the point on the graph is at $y = 4$. Therefore, $f(2) = 4$.

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45 [Challenging] • Target: Order of operations • Cross-Lesson: 6.01 Sequences & Patterns

[illegible] -2 [illegible] What is the number located at the mark -2 on the following number line?

a. -4

b. -1

c. -3

d. **-2** ✓

[illegible] -2. [illegible]. [illegible] -2 [illegible]. The number line shows a marked point at -2. The question asks to identify the value corresponding to this mark. The value is -2 itself.

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